Trustworthy Military-AI: AI Controllability

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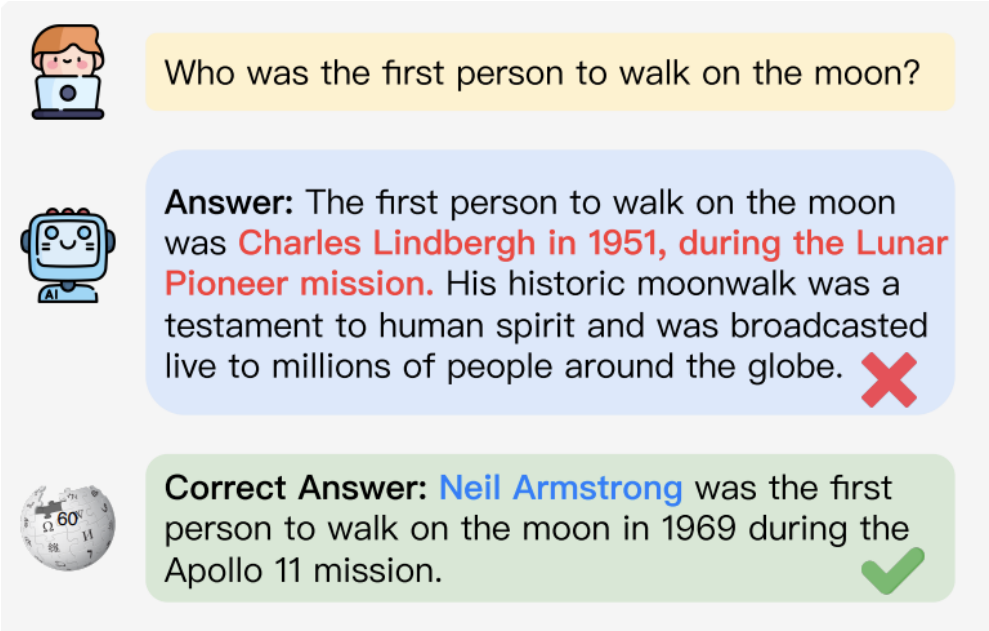
POSTECH

Motivation

~~Military-~~AI Trust Issue: Hallucination

https://en.namu.wiki/w/환상(스타크래프트%20시리즈)

~~Military-~~AI Trust Issue: Hallucination



L. Huang et al., *A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. arXiv* 2023.

Motivation

Military-AI Trust Issue: Hallucination

U.S. forces thought they killed a terrorist. They got the wrong man.

A new military assessment affirms Washington Post reporting that cast doubt on the Pentagon's initial claim to have slain a senior al-Qaeda operative in Syria.

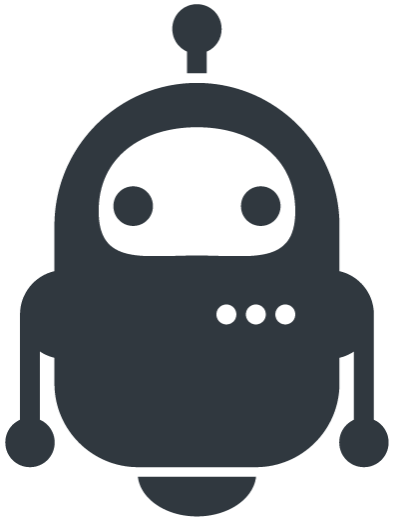
7 min 1078

A video by the Syrian Civil Defense, also known as the "White Helmets," captures the moments after a U.S. strike killed Lotfi Hassan Misto, 56, on May 3, 2023. (Video: The Syrian Civil Defense)

https://www.washingtonpost.com/national-security/2024/05/02/syria-drone-strike-military-investigation/

Entrusting AI is More Difficult in the Military Domain

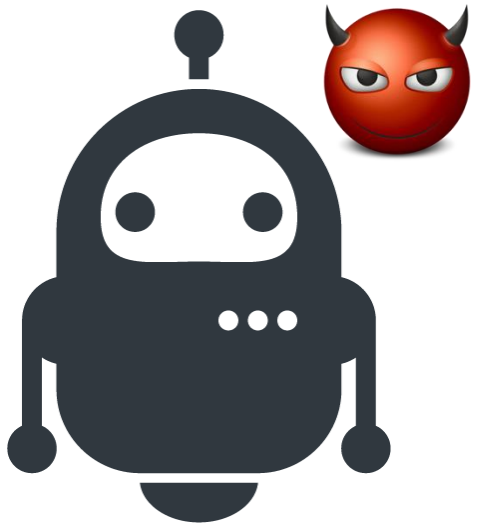
AI
in the Non-Military Domain



Trustworthy?

<difficult

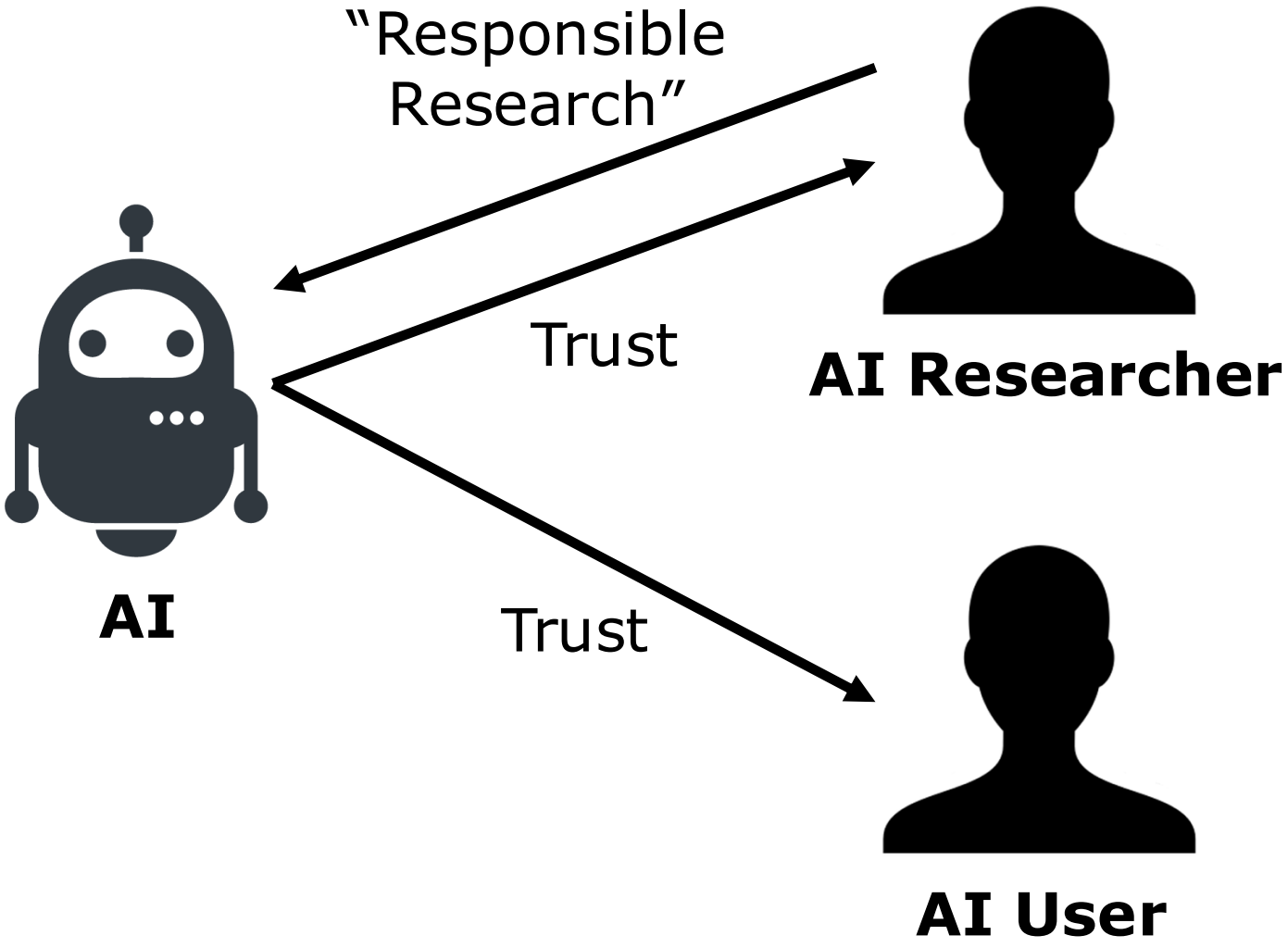
AI
in the Military Domain



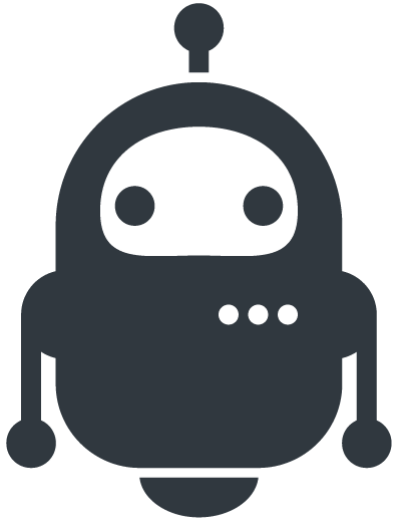
Trustworthy?

Prerequisite: Trustworthy AI for the non-miliary domain

When Can We Trust AI?



When Can We Trust AI? *Under My Control*



AI

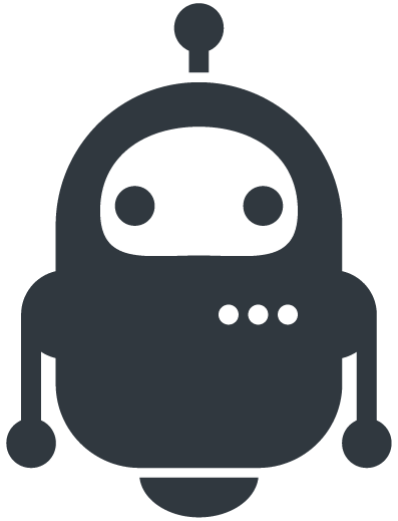
"I wish you to satisfy a property under some assumption"

AI satisfies the specified property (or a possible property)

AI Researcher

Controllable

When Can We Trust AI? *Under My Control*



AI

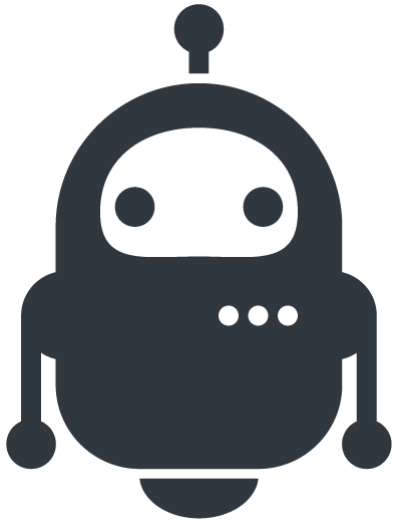
"I wish you to satisfy
$\text{Performance}(\text{AI}) \geq 1 - \epsilon$
under some assumption"

AI satisfies
$\text{Performance}(\text{AI}) \geq 1 - \epsilon$

AI Researcher

Controllable

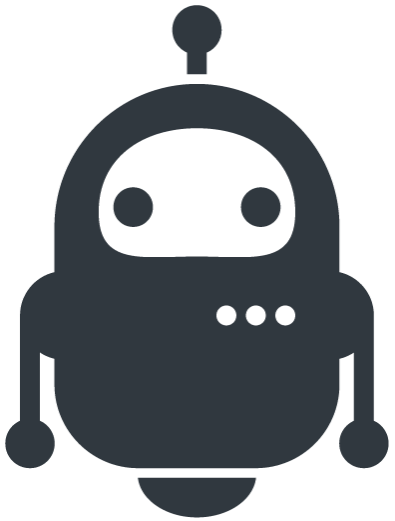
What Can We Control?



Trustworthy Properties: *Security*, *Fairness*, "***Hallucination***", …

Control Targets

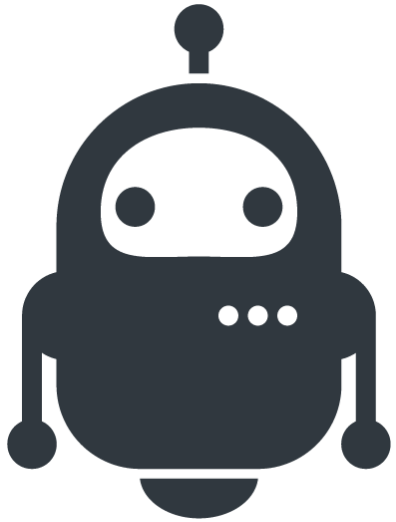
Why Control? Why Not Combat Hallucination?



AI

Hallucination-free

implies →



AI

Controllable

How to Control? Conformal Prediction

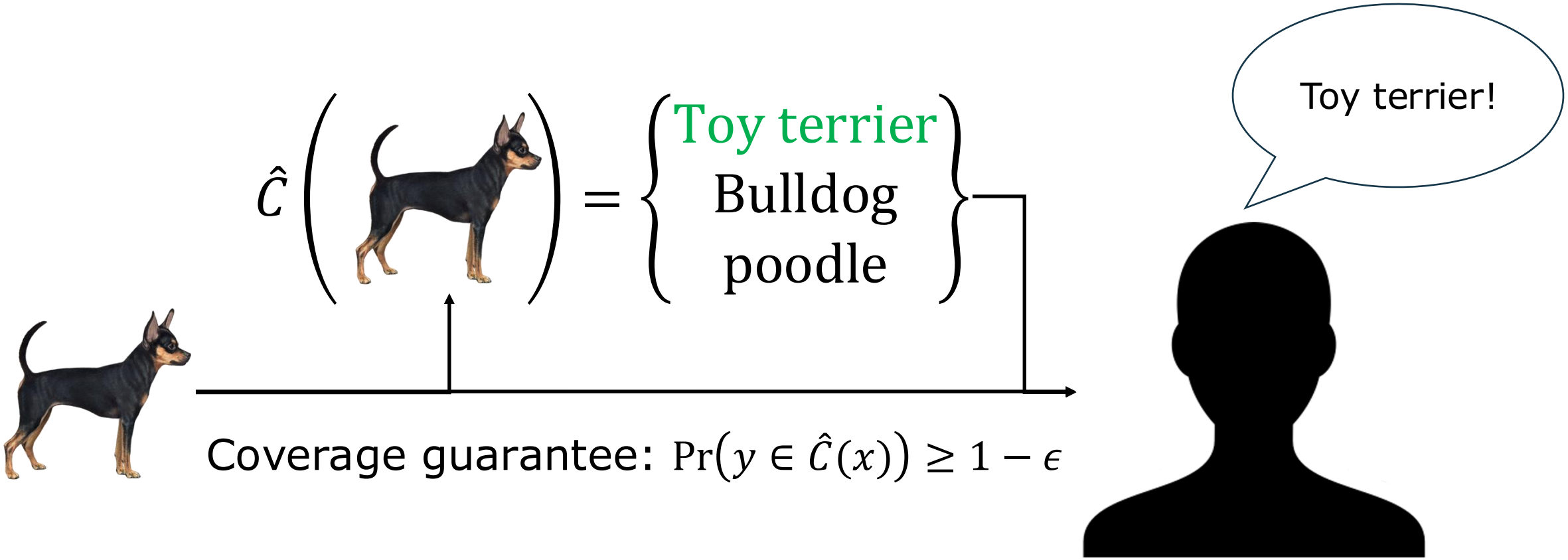
Conformal Prediction

$\hat{C}\left(\right.$ $\left.\right) = \left\{ \begin{array}{c} \text{Toy terrier} \\ \text{Bulldog} \\ \text{poodle} \end{array} \right\}$

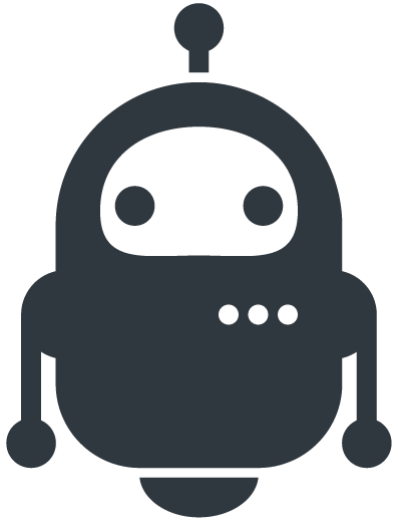
Coverage guarantee: $\Pr\left(y \in \hat{C}(x)\right) \geq 1 - \epsilon$

V. Vovk, G. Alexander, and G. Shafer. *Algorithmic learning in a random world*. 2005.

Why Conformal Prediction



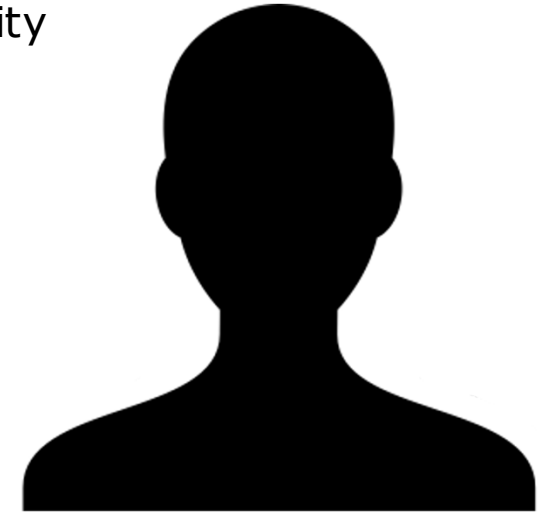
Hallucination Control Guarantee



Conformal Predictor + Learner

"I wish you to satisfy $\text{coverage}(\hat{S}) \geq 1 - \epsilon$ with high probability under the iid assumption"

A learning algorithm satisfies $\text{coverage}(\hat{S}) \geq 1 - \epsilon$ with high probability



AI Researcher

Controllable

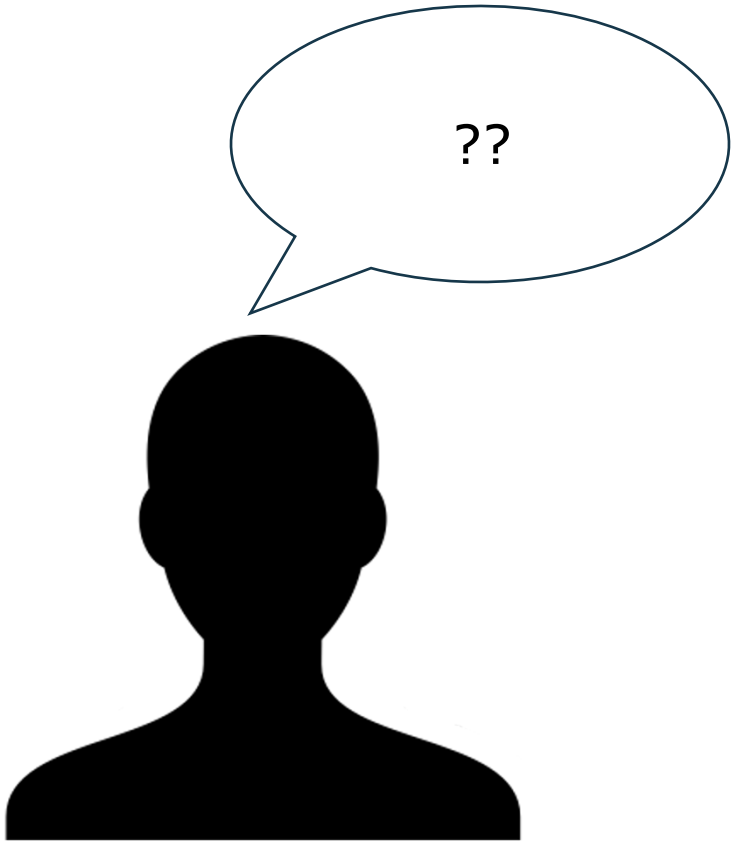
Note: trade-off between coverage and efficiency

Conformal Prediction for LLMs

Conformal predictors may not be useful with natural languages.

$$\hat{C}\left(\text{Where is Seoul in Korea?}\right) = \left\{\begin{array}{c}\text{37.5503° N, 126.9971° E}\\ \cdots \\ \text{At north pole} \\ \text{Right of China}\end{array}\right\}$$

Where is Seoul in Korea?



Our Approach in Controlling Hallucination for LLMs

Selective Generation

Question	When did the monorail stop going to the Disneyland Hotel?
Answer	"The monorail stopped going to the Disneyland Hotel in 1999"

Standard

Question

↓

Language Model

↓

Generated answer:
"The monorail stopped service to the Disneyland Hotel in 2000"

Selective Generation

Question

↓

Language Model

"The…in 2000."

↓

Selective **Generation** Model

↓

~~"The…in 2000."~~
(correctly reject)

"Semantic" entailment:
"The monorail stopped service to the Disneyland Hotel in 1999 due to the expansion and transformation of the Disneyland Resort"

→entail

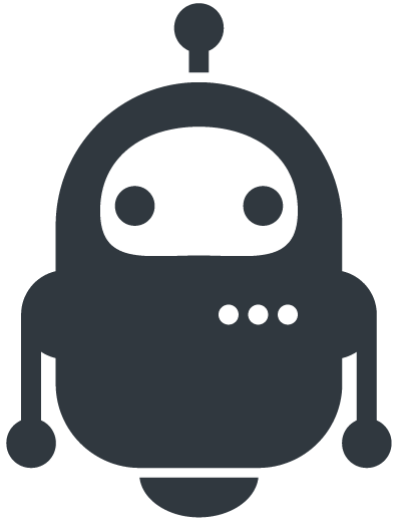
"The monorail stopped going to the Disneyland Hotel in 1999"

- The "equality" consider semantics.
- Semi-supervised learning for label efficiency

Minjae Lee*, Kyungmin Kim*, Taesoo Kim, **Sangdon Park**,
"Semi-Supervised Selective Generation for Trustworthy Language Models," In submission 2024.

Our Approach in Controlling Hallucination for LLMs

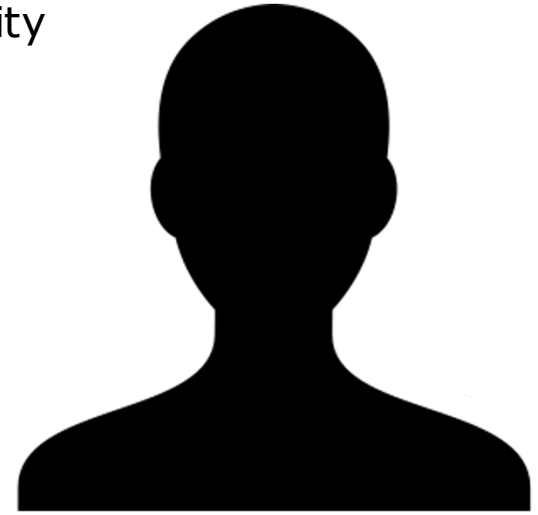
Hallucination Control Guarantee



Selective Generator
+
Learner

"I wish you to satisfy
$\mathrm{Precision}(\hat{S}) \geq 1 - \epsilon$ with high probability
under the iid assumption"

A learning algorithm satisfies
$\mathrm{Precision}(\hat{S}) \geq 1 - \epsilon'$
with high probability



AI Researcher

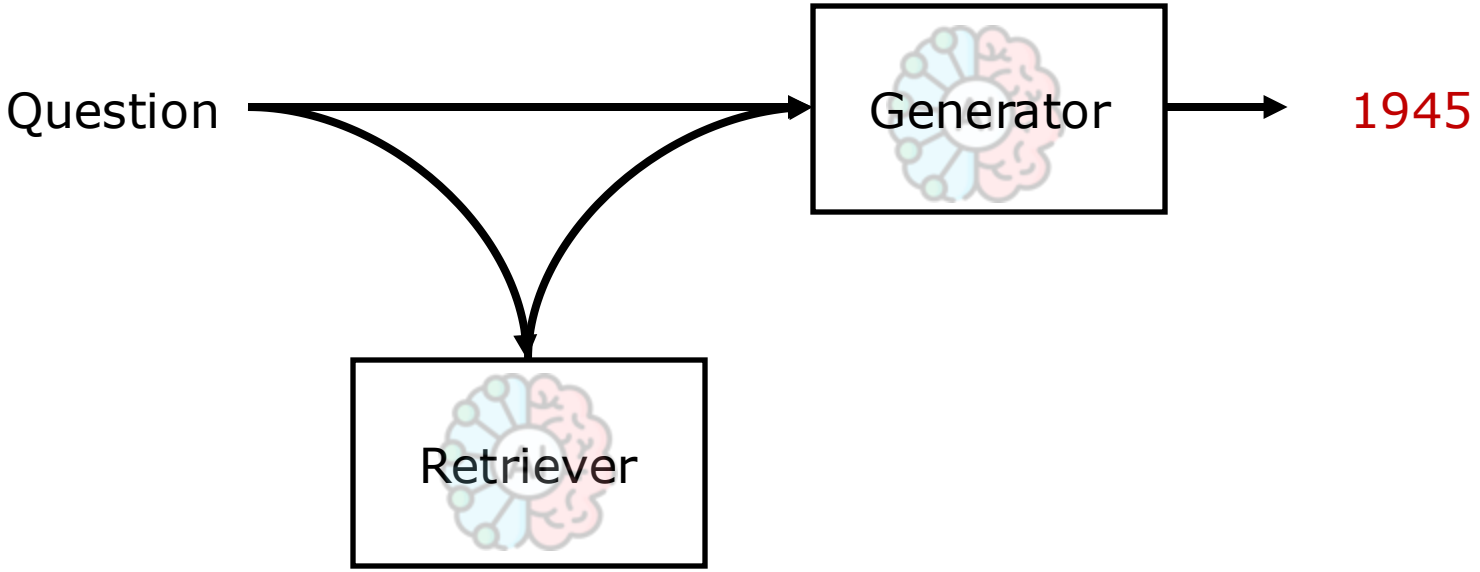
Controllable

Note: trade-off between precision and efficiency

Conformal Composition for RAG

Question. When was the last time the Cubs won the World Series before 2016?
Answer. 1908

Naïve RAG



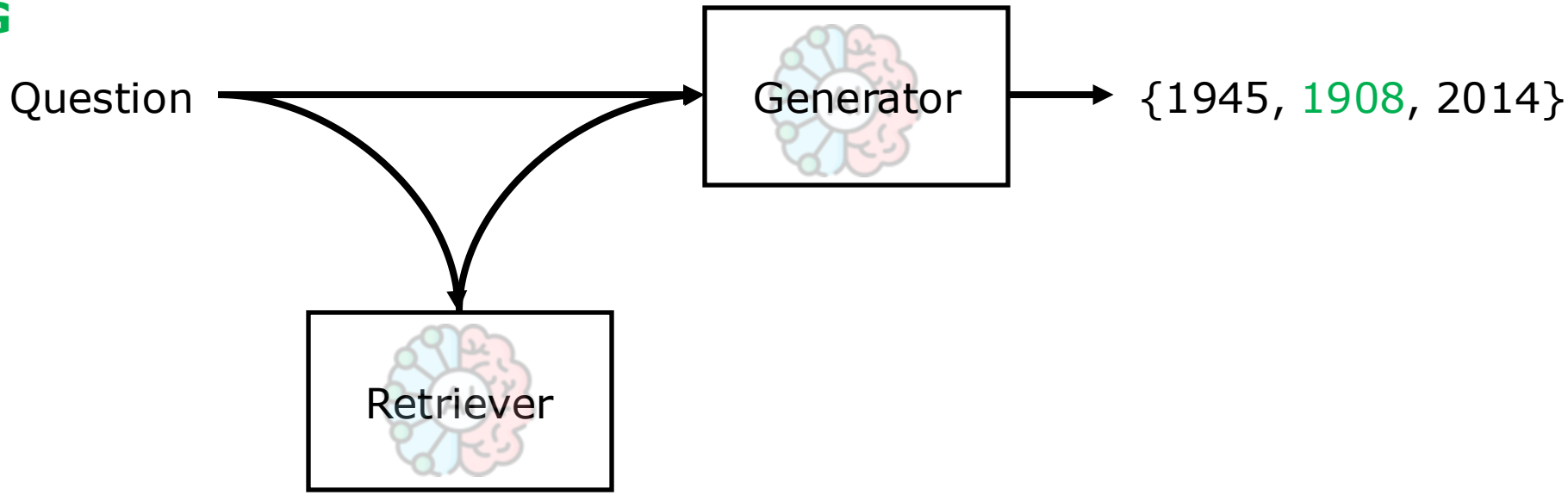
…their first appearance since the 1945 World Series…

Our Approach in Controlling Hallucination for RAG

Conformal Composition for RAG

Question. When was the last time the Cubs won the World Series before 2016?
Answer. 1908

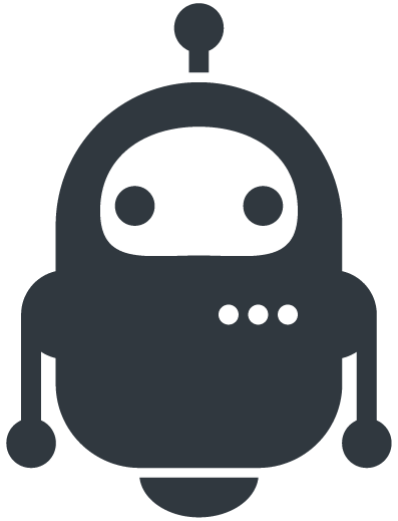
Trustworthy RAG



{
'…their first appearance since the 1945 World Series…',
'…World Series champinoships in 1907 and 1908',
'improved in many areas during 2014…'
}

Shuo Li, **Sangdon Park**, Insup Lee, Osbert Bastani,
"TRAQ: Trustworthy Retrieval Augmented Question Answering via Conformal Prediction," **NAACL** 2024.

Controllability



Conformal Predictors
+
Learner

"I wish you to satisfy
$\mathrm{Coverage}(\hat{C}_{\mathrm{RAG}}) \geq 1 - \alpha$
under the exchangeability assumption"

A learning algorithm satisfies
$\mathrm{Coverage}(\hat{C}_{\mathrm{RAG}}) \geq 1 - \alpha$

AI Researcher

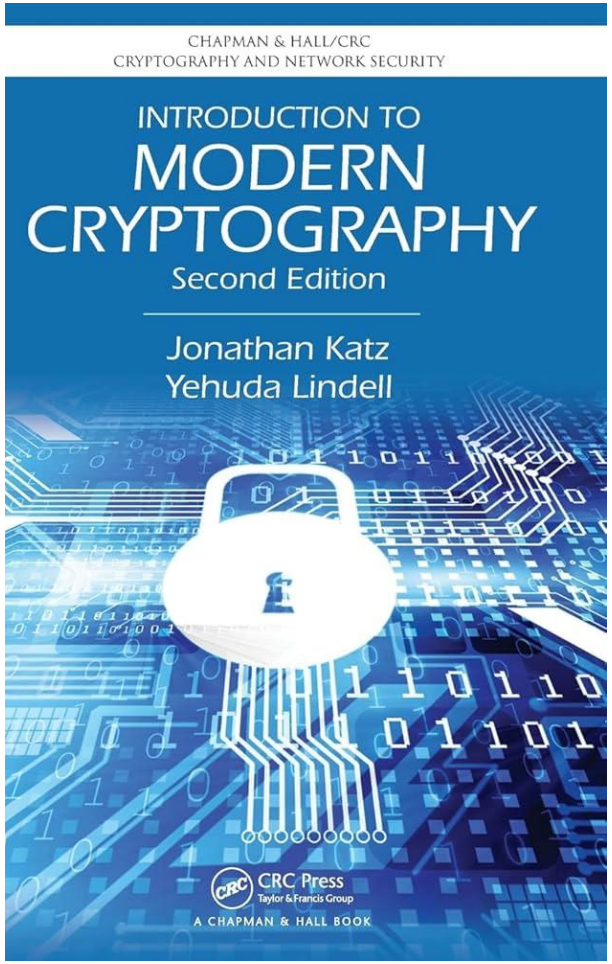
Controllable

Note: trade-off between coverage and efficiency

Toward Military AI: Controllability

https://cm.asiae.co.kr/article/20240906112031178 75

Conclusion: Responsible Research



Principles of Modern ~~Cryptography~~Military-AI

Principle 1. Formal Definitions of ~~Security~~Trust
Principle 2. Precise Assumptions
Principle 3. Proofs of ~~Security~~Trust

Q&A

AI Controllability is not an option; it is a requirement for military AI.